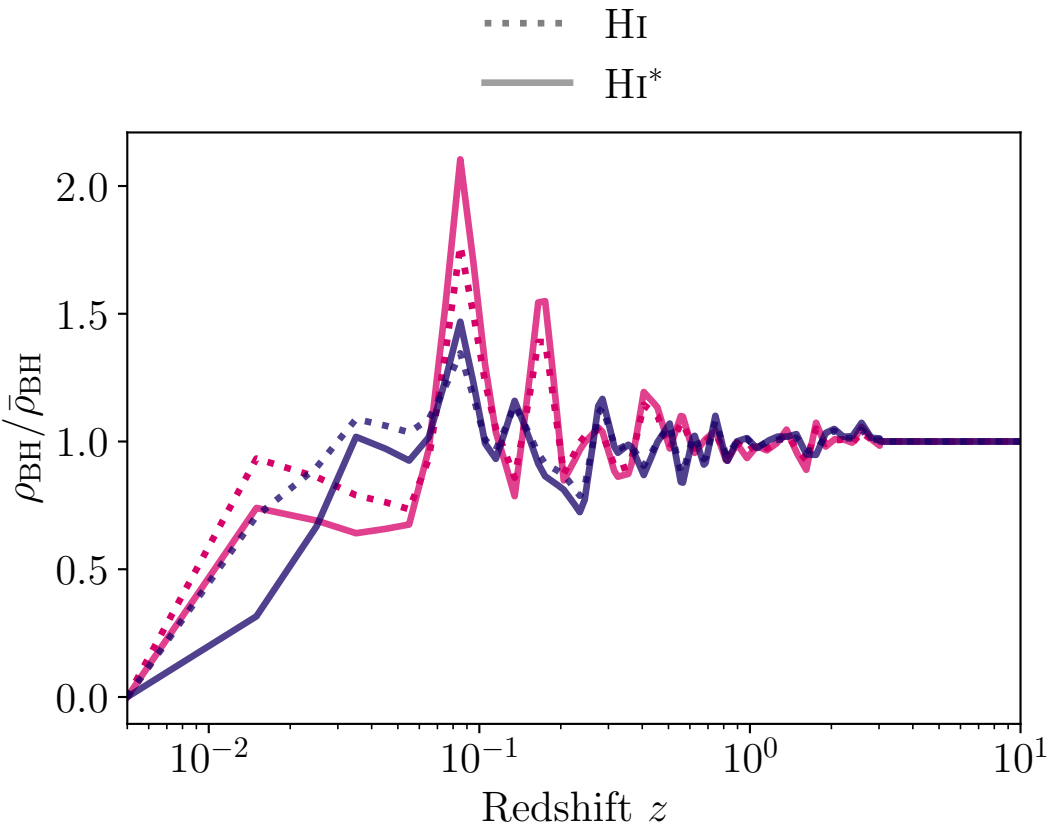

HI
HI*
2.0
1.5
1.0
0.5
0.0
$\rho_{\rm BH}/\bar{\rho}_{\rm BH}$
10^{-2}
10^{-1}
10^{0}
10^{1}
Redshift z